5. REQUIREMENTS TRACEABILITY

Table 5-1 provides the traceability between the X-38 Fault Tolerant Parallel Processor (FTPP) Requirements document number JSC 28671, and this document. The FTPP Requirements document specifies the FTSS system requirements. Table 5-1 is sorted by the system requirement number.

Table 5-1. FTPP to SRS Trace Table.

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
3.1	FTPP System Requirements	Each FTPP shall (3.1.1) consist of five (5) NEs (one for each FCC and one for the NEFU) and FTSS software.	NA
3.1	FTPP System Requirements	For the FTPP system (5 NEs per flight system), the contractor shall (3.1.2) deliver the following end products:	NA
3.1	FTPP System Requirements	The FTPP spare hardware shall (3.1.28) include one (1) radiation hardened FTPP set (5 NEs) and three (3) individual NEs including all optical connects, cables, and required accessories which are flight certified to meet the requirements specified herein for the X-38 space flight vehicles.	NA
3.1	FTPP System Requirements	The contractor shall (3.1.3) develop a preliminary design for the FTPP Architecture.	NA
3.1	FTPP System Requirements	This system shall (3.1.4) provide real time redundancy and fault tolerance among the four FCCs and the	SRS043, SRS091, SRS093, SRS094, SRS095, SRS096, SRS102, SRS104, SRS106, SRS109,

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		NEFU.	SRS128, SRS183, SRS184, SRS187, SRS235, SRS282, SRS283
3.1	FTPP System Requirements	The FTPP system shall not (3.1.5) solely exceed these timing requirement budgets.	SRS034, SRS035
3.1	FTPP System Requirements	In the presence of a maximum 2.5 second power-on skew, the FTPP system shall (3.1.6) be capable of completing FCC system power-up and initialization without synchronization errors.	SRS008
3.1	FTPP System Requirements	Following power being applied to all five chassis, the FTPP system shall (3.1.7) become operational in at most 1.5 minutes.	SRS015
3.1	FTPP System Requirements	The FTPP shall (3.1.26) detect a babbling NE or ICP within 20 milliseconds of the receipt of the first erroneous packet.	SRS235
3.1	FTPP System Requirements	The FTPP shall (3.1.27) recover from a babbling NE or ICP within 40 milliseconds after it is detected.	SRS255
3.1	FTPP System Requirements	An exchange of a single packet of data from an ICP to the FCPs via the NE shall (3.1.8) take no longer than 200 microseconds.	hw
3.1	FTPP System Requirements	An exchange and broadcast of a single packet of data from the FCPs to the ICPs and the FCPs via the NE shall (3.1.9) take	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		no longer than 150 microseconds. A packet size is assumed to be 60 bytes	
3.1	FTPP System Requirements	Under no fault conditions, after initial power on, the FTPP system shall (3.1.10) create six Virtual Groups (VGs), a fault masking FCP, and the five ICPs, and enter the data in the NE Configuration Table (CT).	SRS101, SRS296
3.1	FTPP System Requirements	From the time that the NE failure has been identified, and the NE is recoverable, to the time the NE is recovered shall (3.1.11) be no more than 1.5 minutes.	SRS205
3.1	FTPP System Requirements	The FTPP shall (3.1.12) be capable of restoring a corrected faulty computer into the flight critical computer set.	SRS124, SRS214
3.1	FTPP System Requirements	The FTPP shall (3.1.14) take no more than 1 second per Megabyte of data to be realigned.	SRS203
3.1	FTPP System Requirements	The failed channel, provided it is recoverable, shall (3.1.15) be recovered in less than 1.5 minutes.	SRS205
3.1	FTPP System Requirements	The FTPP voting implementation shall (3.1.16) isolate and remove a faulty computer from the flight critical computer set within 60 milliseconds, once the	SRS109

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		fault has manifested itself.	
3.1	FTPP System Requirements	The FTPP system shall (3.1.17) be two fault tolerant for any two non-simultaneous hardware faults through out all phases of the X-38 mission (i.e., from power on to power off, without degradation).	SRS043, SRS091, SRS093, SRS094, SRS095, SRS096, SRS102, SRS104, SRS106, SRS109, SRS128, SRS183, SRS184, SRS187 SRS235, SRS282, SRS283, SRS281
3.1	FTPP System Requirements	The FTPP system shall (3.1.18) be capable of powering up and operating with any combination of 3 of the 5 FCR's active.	SRS010, SRS102
3.1	FTPP System Requirements	The FTPP system shall (3.1.19) be able to accommodate power up of all 5 channels and maintain all 5 NEs active, assuming no failures.	SRS201
3.1	FTPP System Requirements	The FTPP system shall (3.1.20) incorporate additional channels in the active set as they are powered-on by the application software.	SRS124, SRS125, SRS126, SRS214
3.1	FTPP System Requirements	After the initial power-up of only two FCRs and the NEFU, the FTPP system shall (3.1.25) be able to incorporate two more FCRs, upon their simultaneous or separate power-up.	SRS236
3.1	FTPP System Requirements	The FTPP system shall (3.1.21) monitor for the presence of new channels at a 1 Hz periodic rate.	SRS124
3.1	FTPP System Requirements	All healthy FCRs shall (3.1.22) be	SRS125, SRS126, SRS214

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		incorporated as they become available.	
3.1	FTPP System Requirements	All FCP processing channels (up to 4) shall (3.1.23) be incorporated into the FCP virtual group as they become available, provided that recovery and memory alignment is allowed by the application software.	SRS123, SRS124, SRS125, SRS126, SRS214
3.1	FTPP System Requirements	When memory realignment is not permitted, the FTPP system shall (3.1.24) maintain, at a minimum, 3 channels of I/O or 2 channels of I/O and the NEFU.	SRS254
3.2.1	Network Element Addressing Convention	The FCC hardware shall (3.2.1.1) use 3-digit binary numbers as outlined above for both addresses.	hw
3.2.2.1	Data Exchange Primitives	The X-38 NE shall (3.2.2.1.1) provide four types of data exchange primitives as described below, in accordance with the Byzantine-resilient replicated determinism requirements described in [1]. (Class 0, 1, 2, Broadcast)	hw
3.2.2.2	Configuration Table Updates	The NE shall (3.2.2.2.1) keep track of the grouping of physical processors into virtual groups. This mapping is contained in the Configuration Table (CT).	hw
3.2.2.2	Configuration Table Updates	The CT shall (3.2.2.2.2) also	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		contain time-outs and vote masks.	
3.2.2.2	Configuration Table Updates	It shall (3.2.2.2.3) be possible to modify the CT whenever any of this information is changed by using a CT update primitive in a synchronous and atomic manner.	hw
3.2.2.3	Initial Synchronization (ISYNC)	The NE shall (3.2.2.3.1) be configured to automatically enter ISYNC microcode following power on.	hw
3.2.2.3	Initial Synchronization (ISYNC)	The NE shall (3.2.2.3.2) start transmitting a sync message using class 2 exchanges once 3 fault tolerant clocks have synchronized, thus enabling inter-NE data exchanges.	hw
3.2.2.3	Initial Synchronization (ISYNC)	A time-out period shall (3.2.2.3.3) be started when 3 NEs have joined in.	hw
3.2.2.3	Initial Synchronization (ISYNC)	The ISYNC procedure shall (3.2.2.3.4) terminate when all 5 NEs are synchronized, four NEs are synchronized (when only four NEs are powered), or after the time-out period.	hw
3.2.2.3	Initial Synchronization (ISYNC)	The NE microcode shall (3.2.2.3.6) initialize the time-outs and vote masks in the NE CT using values stored in the NE.	hw
3.2.2.3	Initial Synchronization (ISYNC)	The maximum time-out value shall (3.2.2.3.7) be 327.68 microseconds (256	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		counts of the least significant bit of the fault tolerant clock which is 1.28 microseconds).	
3.2.2.4	Transient NE Recovery (TNR)	An NE that fails to synchronize with other NEs during ISYNC after a pre-defined time-out period shall (3.2.2.4.1) then enter TNR microcode.	hw
3.2.2.4	Transient NE Recovery (TNR)	An NE shall (3.2.2.4.12) directly enter TNR microcode after a voted reset or an NE watchdog timer reset.	hw
3.2.2.4	Transient NE Recovery (TNR)	It shall (3.2.2.4.2) stay in TNR mode indefinitely until a successful TNR exchange is observed.	hw
3.2.2.4	Transient NE Recovery (TNR)	The operational NEs shall (3.2.2.4.3) enter a "working group" TNR routine when the FCPs request to send a TNR packet.	hw
3.2.2.4	Transient NE Recovery (TNR)	If no new NE is observed, the functioning NEs shall (3.2.2.4.6) return to the operational mode within 500 microseconds.	hw
3.2.2.4	Transient NE Recovery (TNR)	If there is a new NE, the state of the reintegrated NE shall (3.2.2.4.13) be made congruent with the state of the operational NEs as follows.	hw
3.2.2.4	Transient NE Recovery (TNR)	The Configuration Table shall (3.2.2.4.7) be exchanged and voted into the newly	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		recovered NE.	
3.2.2.4	Transient NE Recovery (TNR)	Time-outs in the scoreboard shall (3.2.2.4.8) be aligned by resetting all time-outs.	hw
3.2.2.4	Transient NE Recovery (TNR)	The global synchronous timer shall (3.2.2.4.9) be realigned by exchanging and voting the timer value. The timer value will stop incrementing until the realignment of the timer is complete.	hw
3.2.2.4	Transient NE Recovery (TNR)	Provided the failed NE is in a recoverable state, recovery of a failed NE shall (3.2.2.4.11) take no longer than 500 microseconds. This recovery requirement includes the time from which the FTSS software initiates the recovery to the time the recovery is complete.	hw
3.2.2.5	Voted Resets	There shall (3.2.2.5.1) be built-in support on the NE for voted resets, including a special packet type for executing the primitive.	hw
3.2.2.5	Voted Resets	The NE shall (3.2.2.5.3) provide the capability to perform a voted VME bus reset.	hw
3.2.2.6	Error Syndrome Reports	The NE shall (3.2.2.6.1) place error syndromes in the input information block whenever a packet is successfully delivered to the FCP.	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
3.2.2.6	Error Syndrome Reports	The NE syndromes shall (3.2.2.6.2) be located in the second longword of the input information block buffer cell.	hw
3.2.2.6	Error Syndrome Reports	The NE syndromes shall (3.2.2.6.3) include indications of vote errors, fault-tolerant clock synchronization errors, and fiber-optic link errors, as detailed below.	hw
3.2.2.6	Error Syndrome Reports	Each syndrome shall (3.2.2.6.4) represent an occurrence of the indicated error at some time between delivery of the previous packet and delivery of the current packet.	hw
3.2.2.6	Error Syndrome Reports	The scoreboard syndromes shall (3.2.2.6.5) be located in the third longword of the input information block buffer cell.	hw
3.2.2.6	Error Syndrome Reports	They shall (3.2.2.6.6) include indications of scoreboard vote errors, Output Buffer Not Empty (OBNE) time-outs, and Input Buffer Not Full (IBNF) time-outs, as follows.	hw
3.2.2.6	Error Syndrome Reports	When a majority, but not a unanimity, of FCP members are observed with packets in their output buffers, a time-out shall (3.2.2.6.7) be initiated.	hw
3.2.2.6	Error Syndrome Reports	If the time-out expires before the other members transmit the	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		packet, the remaining member shall (3.2.2.6.8) be ignored, the packet exchanged, and an OBNE time-out recorded.	
3.2.2.6	Error Syndrome Reports	When a majority, but not a unanimity, of FCP members are observed with room in their input buffers, a time-out shall (3.2.2.6.15) be initiated.	hw
3.2.2.6	Error Syndrome Reports	If the time-out expires before the other members have room in their input buffers, any member without room in their input buffer shall (3.2.2.6.16) be ignored, the packet exchanged, and an IBNF time-out recorded.	hw
3.2.2.7	Timestamps	The NE shall (3.2.2.7.1) place a timestamp in the input information block of each packet successfully delivered to an FCP or ICP.	hw
3.2.2.7	Timestamps	The timestamps shall (3.2.2.7.2) be congruent across all members of the destination FCP.	hw
3.2.2.7	Timestamps	The timestamps shall (3.2.2.7.3) also be congruent across all active processors in the case of a broadcast.	hw
3.2.2.7	Timestamps	The timestamp shall (3.2.2.7.4) be a 32-bit quantity that indicates relative time within the FCC.	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
3.2.2.7	Timestamps	The resolution of the timestamp shall (3.2.2.7.5) be 1.28 microseconds.	hw
3.2.2.7	Timestamps	When the timestamp counter reaches the maximum value (Hex FFFFFFFF or approximately 5500 seconds), it shall (3.2.2.7.6) wrap around to zero.	hw
3.2.2.7	Timestamps	The timestamp counter shall (3.2.2.7.7) be initialized to zero during ISYNC.	hw
3.2.2.7	Timestamps	The counter shall (3.2.2.7.8) increase monotonically after that, except during TNR.	hw
3.2.2.7	Timestamps	The timestamps shall (3.2.2.7.9) be frozen during TNR until the realignment is complete.	hw
3.2.2.8	Debug Commands	The NE shall (3.2.2.8.1) implement in microcode support commands to aid in debugging new NEs and for performing stand-alone diagnostics and self-tests.	hw
3.2.2.8	Debug Commands	As a minimum, the following diagnostic functionality shall (3.2.2.8.2) be supported. 1. VMEbus Interface test 2. Message wraparound test 3. Class 1 Voter test 4. Class 2 Voter test 5. Input Buffer Test 6. Output Buffer test	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		7. CT entry test 8. CT update test 9. Timestamp test	
3.2.3.1.1	Prototype NE Physical Characteristics	The prototype NE shall (3.2.3.1.1.1) reside on a single commercial grade 6U VME board.	hw
3.2.3.1.1	Prototype NE Physical Characteristics	The prototype NE shall (3.2.3.1.1.2) dissipate no more than 35 Watts.	hw
3.2.3.1.1	Prototype NE Physical Characteristics	The operating temperature range for the prototype NEs shall (3.2.3.1.1.3) be from 0 to 32.2º C at the inlet to the cooling fans.	hw
3.2.3.1.1	Prototype NE Physical Characteristics	The storage temperature range shall (3.2.3.1.1.4) be from - 30º to + 60º C.	hw
3.2.3.1.1	Prototype NE Physical Characteristics	The prototype NEs shall (3.2.3.1.1.5) use convection cooling.	hw
3.2.3.1.1	Prototype NE Physical Characteristics	The prototype NE shall (3.2.3.1.1.6) be fabricated using commercial grade components.	hw
3.2.3.1.2	Flight NE Physical Characteristics	Each flight NE shall (3.2.3.1.2.1) reside on a single ruggedized, wedge-locked, conduction-cooled 6U VME board.	hw
3.2.3.1.2	Flight NE Physical Characteristics	It shall (3.2.3.1.2.2) be able to be installed in an Air Transportable Rack (ATR) Chassis with .8 pitch spacing card slots.	hw
3.2.3.1.2	Flight NE Physical Characteristics	The conduction-cooled boards' mechanical core shall (3.2.3.1.2.3) be designed in	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		accordance with IEEE 1101.2.	
3.2.3.1.2	Flight NE Physical Characteristics	Power de-coupling mechanisms shall (3.2.3.1.2.4) be designed into the NE.	hw
3.2.3.1.2	Flight NE Physical Characteristics	Backplane connector form factors shall (3.2.3.1.2.5) be in accordance with the VME64 with extensions (5 row P1 and P2) draft standard.	hw
3.2.3.1.2	Flight NE Physical Characteristics	Connector P2 shall (3.2.3.1.2.6) have all pins in row z connected to ground;Row d pins 1 through 31 connected to ground and row d pin 32 connected to +5 VDC.	hw
3.2.3.1.2	Flight NE Physical Characteristics	Connector P1 shall (3.2.3.1.2.7) have Row z pins 1 through 32 connected to ground, row d pins 1 and 32 connected to +5 VDC, row d pins 2 through 31 connected to ground with the exception of pins 3 through 8, pins 12, 14, 16, 18, 20, 22, 24, 26, 28, and 30 which will not be connected.	hw
3.2.3.1.2	Flight NE Physical Characteristics	Connector P1 pins 1 and 32 of row d shall (3.2.3.1.2.8) be connected to +5V.	hw
3.2.3.1.2	Flight NE Physical Characteristics	The inter-NE communications shall (3.2.3.1.2.9) be through fiber.	hw
3.2.3.1.2	Flight NE Physical Characteristics	Test points shall (3.2.3.1.2.10) be made available at the	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		P2 connector for examining the operational status of the NE.	
3.2.3.1.2	Flight NE Physical Characteristics	Each flight NE shall (3.2.3.1.2.11) dissipate no more than 35 Watts.	hw
3.2.3.1.2	Flight NE Physical Characteristics	Each flight NE shall (3.2.3.1.2.12) be conduction-cooled.	hw
3.2.3.1.2	Flight NE Physical Characteristics	Flight hardware shall (3.2.3.1.2.13) be fabricated using radiation-hardened and/or radiation tolerant components.	hw
3.2.3.1.2	Flight NE Physical Characteristics	Fabrication and assembly of the boards shall (3.2.3.1.2.14) meet NAS 5300.4(3L), NAS 5300.4(3J-1), NHB5300.4(3A-2), IPC 275, IPC 6011, IPC 6012 and GSFC-S-312-P-003.	hw
3.2.3.1.3	Flight NE Environmental Qualification Conditions	The flight NE shall (3.2.3.1.3.1) be capable of meeting all performance requirements specified herein during and after exposure to the environmental service conditions specified herein.	hw
3.2.3.1.3	Flight NE Environmental Qualification Conditions	The flight NE shall (3.2.3.1.3.2) be designed and constructed so that no part of any component shifts in setting, position, or adjustment.	hw
3.2.3.1.3	Flight NE Environmental Qualification	No degradation shall (3.2.3.1.3.3) be caused in the	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
	Conditions	performance that is specified in Subsections 3.2.3.1.3.1 through 3.2.3.1.3.11	
3.2.3.1.3.1	Temperature	The flight NE shall (3.2.3.1.3.1.1) meet the following temperature requirement while operating, 32 F to 149 F (0 C to 65 C) at the card edge.	hw
3.2.3.1.3.1	Temperature	The NE shall (3.2.3.1.3.1.2) operate with a worst case card edge thermal interface temperature of +65 °C for greater than 10 hours.	hw
3.2.3.1.3.1	Temperature	The storage (i.e., non-operating) temperature range for the flight NE shall (3.2.3.1.3.1.3) be from -30 °C to 60 °C.	hw
3.2.3.1.3.1	Temperature	For qualification testing, the flight NE shall (3.2.3.1.3.1.4) meet the following temperature requirement while operating, 12 F to 152 F (-11 C to 66.7 C) at the card edge.	hw
3.2.3.1.3.2	Vibration	The flight NE shall (3.2.3.1.3.2.1) be capable of complying with all of the performance specified herein while not operating during all specified levels.	hw
3.2.3.1.3.2.1	Random Vibration	The flight NE shall (3.2.3.1.3.2.1.1) be capable of withstanding the following environment in non-operational	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		mode: Frequency, Hz Qualification Level g2/Hz 20 0.026 20-50 +3 dB/Octave 50-800 0.16 800-2000 -3 dB/Octave 2000 0.026	
3.2.3.1.3.2.1	Random Vibration	For qualification testing, the sweep time per axis shall (3.2.3.1.3.2.1.2) be 3 minutes, 14.1 g-rms overall.	hw
3.2.3.1.3.2.1	Random Vibration	The flight NE shall (3.2.3.1.3.2.1.3) be non operating during the application of this vibration in all axes.	hw
3.2.3.1.3.3	Ionization Radiation	The flight NE shall (3.2.3.1.3.3.1) be designed to be capable of complying with all the performance requirements specified herein while being subjected to the total dose, single event upset and latchup immune requirements in SSP 30512 Rev. C.	hw
3.2.3.1.3.3	Ionization Radiation	All radiation test results shall (3.2.3.1.3.3.7) be characterized and documented.	hw
3.2.3.1.3.4	Shock (Non-operating)	The flight NE shall (3.2.3.1.3.4.1) be designed to be capable of complying with all the performance requirements specified herein after being subjected to a total of	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		six impact shocks, consisting of six shocks (one in each opposite direction) along each of three orthogonal axes.	
3.2.3.1.3.4	Shock (Non-operating)	The waveform and amplitude of the shock pulses shall (3.2.3.1.3.4.2) be sawtooth shock pulse 20g peak, 11 milliseconds nominal duration.	hw
3.2.3.1.3.5	Humidity	The flight NE shall (3.2.3.1.3.5.1) be designed to comply with all of the performance requirements specified herein while withstanding the effects of up to 90% relative humidity, non condensing while operating.	hw
3.2.3.1.3.6	Pressure	The flight NE shall (3.2.3.1.3.6.1) be designed to be capable of complying with all the performance requirements specified herein while non-operating in an atmosphere of 8 to 18 Psia.	hw
3.2.3.1.3.8	Electromagnetic Radiation	The flight NE shall (3.2.3.1.3.8.1) be designed to be electromagnetically compatible with the Radstone Power PC 604R	hw
3.2.3.1.3.10.1	MTBF	The design, manufacturing, and radiation environment composite failure rate/Mean Time	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		Between Failures (MTBFs) of the flight NE shall (3.2.3.1.3.10.1.1) be predicted using techniques in MIL-HDBK 217 or other commonly accepted procedures.	
3.2.3.1.3.10.2	Operational Service Life	The useful operating service life shall (3.2.3.1.3.10.2.1) be a minimum of 30,000 hours.	hw
3.2.3.1.3.10.2	Operational Service Life	The useful operating life shall (3.2.3.1.3.10.2.2) be determined by analysis and presented at the critical design review.	hw
3.2.3.1.3.10.3	Storage Life	The flight NE storage life shall (3.2.3.1.3.10.3.1) be 5 years or better.	hw
3.2.3.1.3.12.1	Temperature Range	For acceptance testing, the flight NE shall (3.2.3.1.3.12.1.3) meet the following temperature requirement while operating, 32 F to 132 F (0 C to 55.6 C) at the card edge.	hw
3.2.3.1.3.12.2	Random Vibration Requirements	The ESS random vibration Power Spectral Density for the flight NE shall (3.2.3.1.3.12.2.1) be as follows: Total 10.8 grms …[Frequency, Hz Level g2/Hz 20 0.015 20-50 +3 dB/Octave 50-800 0.09 800-2000 -3 dB/Octave	hw

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		2000 0.015	
3.2.3.2	Operational Characteristics	The X-38 NE shall (3.2.3.2.1) be able to communicate with 4 other NEs.	hw
3.2.3.2	Operational Characteristics	The NE shall (3.2.3.2.2) support at least two physical processors per FCR.	hw
3.2.3.2	Operational Characteristics	The NE shall (3.2.3.2.3) support at least 6 virtual groups.	hw
3.2.3.2	Operational Characteristics	The NE shall (3.2.3.2.5) support class 1 and class 2 bandwidth of at least 1 Mbyte/sec.	hw
3.2.3.2	Operational Characteristics	The skew between NEs, measured at delivery of messages to dual-port RAM accessible by the processors via the VME bus, shall (3.2.3.2.6) be no more than 100 nsecs.	hw
3.2.3.2	Operational Characteristics	The NEs shall (3.2.3.2.7) be able to achieve phase-locked synchronization of the fault-tolerant clocks in less than 5 milliseconds from the time the last FCC powered up and exited the reset state	hw
3.2.4	Miscellaneous NE Requirements	The contractor shall (3.2.4.1) maintain a product identification and tracking system.	NA
3.2.4	Miscellaneous NE Requirements	Each NE and flight cable assembly shall (3.2.4.2) be identified by a part or type number and a unique serial number, consistent with the configuration	NA

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		management system and the specification for the contract.	
3.2.5	Network Element Fifth Unit (NEFU) Requirements	The presence, or absence of, an NEFU ICP shall (3.2.5.2) not impact the NE firmware (i.e., the NE firmware will not be different).	hw
3.3.1	Programming Language and Operating System [2]	Fault Tolerant System Services (FTSS) software shall (3.3.1.1) use VxWorks Version 5.4 as the Operating System.	SRS160
3.3.1	Programming Language and Operating System [2]	The software shall (3.3.1.2) be written in the C programming language, with the exception of the system loader software which may be written in scripts, and operate on a PowerPC 604R.	SRS158, SRS159
3.3.1	Programming Language and Operating System [2]	The FTSS software and the VxWorks operating system, together, shall (3.3.1.3) take up no more than 3 Megabytes of ROM, when loaded into FCP or ICP memory.	SRS193
3.3.1	Programming Language and Operating System [2]	The FTSS software and the VxWorks operating system shall (3.3.1.4) utilize no more than 9 Megabytes of DRAM code and data space.	SRS258
3.3.1	Programming Language and Operating System [2]	Of the 9 Megabytes of DRAM allocation, only 4 Megabytes of FTSS/VxWork's DRAM shall (3.3.1.5) be re-aligned during any re-alignment	SRS259

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		attempts.	
3.3.2	Start Up	Upon CPU reset caused by power on, watchdog timer or by other means, Start Up shall (3.3.2.1) execute the initial BIT (IBIT).	SRS237
3.3.2	Start Up	After successfully completing IBIT, the software shall (3.3.2.3) continue with the initialization of VxWorks and FTSS software.	SRS234
3.3.2	Start Up	If MPCC IBIT failed, the FTSS SW shall (3.3.2.24) switch to using the redundant MPCC card in that C&T FCR.	SRS299
3.3.2	Start Up	If the 5th ICP fails, the FTSS SW shall (3.3.2.25) ignore the error and allow the NE to continue the synch process and become part of the voting group, if possible.	SRS243
3.3.2	Start Up	On each FCP, the FTPP system shall (3.3.2.26) configure the Radstone firmware to perform the IBIT tests shown in Table 3.3.2.1, FCP IBIT Table.	SRS260
3.3.2	Start Up	The FTPP system shall (3.3.2.27) configure the Radstone processor to halt processing if any of the MPE tests, mentioned in Table 3.3.2.1, FCP IBIT Table, fail.	SRS261
3.3.2	Start Up	The FTPP system shall (3.3.2.28) configure the	SRS262

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		Radstone processor to continue processing if any of the Power-up tests or Initial BIT tests mentioned in Table 3.3.2.1, FCP IBIT Table, fail.	
3.3.2	Start Up	In all FCP IBIT cases, provided the hardware state permits, the FTSS shall (3.3.2.29) log the error and report it in the X-38 telemetry stream.	SRS239
3.3.2	Start Up	Upon completion of logging a Power-up test or Initial BIT test failure, the FTSS system shall (3.3.2.43) consider the FCP failed and attempt recovery actions as stated in requirement 3.3.5.29.	SRS290
3.3.2	Start Up	On each ICP, the FTPP system shall (3.3.2.30) configure the Radstone firmware to perform the IBIT tests shown in Table 3.3.2.2, ICP IBIT Table.	SRS287
3.3.2	Start Up	The FTPP system shall (3.3.2.31) configure the Radstone processor to halt processing if any of the MPE tests, mentioned in Table 3.3.2.2, ICP IBIT Table, fail.	SRS288
3.3.2	Start Up	The FTPP system shall (3.3.2.32) configure the Radstone processor to continue processing if any of the Power-up tests or Initial BIT tests mentioned in	SRS289

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		Table 3.3.2.2, ICP IBIT Table, fail.	
3.3.2	Start Up	In all ICP IBIT cases, provided the hardware state permits, the FTSS shall (3.3.2.33) log the error and report it in the X-38 telemetry stream.	SRS239
3.3.2	Start Up	Upon completion of logging a Power-up test or Initial BIT test failure, the FTSS system shall (3.3.2.44) consider the ICP failed and attempt recovery actions as stated in requirement 3.3.5.29.	SRS290
3.3.2	Start Up	On each ICP/PMC1553, the FTPP system shall (3.3.2.34) configure the Radstone firmware to perform the IBIT tests shown in Table 3.3.2.3, ICP/PMC1553 IBIT Table.	SRS264
3.3.2	Start Up	The FTPP system shall (3.3.2.35) configure the Radstone ICP/PMC1553 card to halt processing if any of the MPE tests, mentioned in Table 3.3.2.3, ICP/PMC1553 IBIT Table, fail.	SRS265
~~3.3.2~~	~~Start Up~~	~~The FTPP system shall (3.3.2.36) configure the Radstone ICP/PMC1553 card processor to continue processing if any of the Power-up tests or Initial BIT tests mentioned in Table 3.3.2.3, ICP/PMC1553 IBIT Table, fail.~~	~~SRS266~~

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
3.3.2	Start Up	The FTPP system shall (3.3.2.36) configure the Radstone ICP/PMC1553 card to continue processing if any of the Initial BIT tests mentioned in Table 3.3.2.3, ICP/PMC1553 IBIT Table, fail.	SRS266
3.3.2	Start Up	In all ICP/PMC1553 IBIT cases, provided the hardware state permits, the FTSS shall (3.3.2.37) log the error and report it in the X-38 telemetry stream.	SRS239
3.3.2	Start Up	On each MPCC, the FTPP system shall (3.3.2.38) configure the Radstone firmware to perform the IBIT tests shown in Table 3.3.2.4, MPCC IBIT Table.	SRS267
3.3.2	Start Up	In all MPCC IBIT cases, provided the hardware state permits, the FTSS shall (3.3.2.40) log the error and report it in the X-38 telemetry stream.	SRS239
3.3.2	Start Up	If IBIT fails, the FTSS SW shall (3.3.2.41) handle the failure as stated in the preceding IBIT requirements and in requirement 3.3.5.29's table, FTPP Failure Response/Recovery Mechanisms.	SRS269
3.3.2	Start Up	However, the FTSS SW shall (3.3.2.42) notify the application software if the 5th	SRS098

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		ICP's heartbeat ceases to exist.	
3.3.2	Start Up	The surviving triplex shall (3.3.2.5) attempt to sync with the failed FCP.	SRS177
3.3.2	Start Up	If the failed FCP has not synced in 2.5 seconds, after the surviving triplex has detected the loss of the FCP, then the surviving triplex shall (3.3.2.6) send a voted VMEbus reset through the NE to the failed FCP.	SRS178
3.3.2	Start Up	Start Up shall (3.3.2.12) synchronize its FCP with other operational FCPs.	SRS008
3.3.2	Start Up	Start Up shall (3.3.2.13) make their state congruent.	SRS011
3.3.2	Start Up	The FCP state shall (3.3.2.14) include all volatile memory, read/write memory, registers, timers, and counters, except that part of the memory exclusively set aside for channel-unique information.	SRS011
~~3.3.2~~	~~Start Up~~	~~It shall (3.3.2.17) support normal synchronization following power on or reset~~	~~SRS194~~
3.3.2	Start Up	It shall (3.3.2.17) support normal synchronization following power on or reset.	SRS194
3.3.2	Start Up	Start Up shall (3.3.2.18) be able to synchronize all operational FCPs in	SRS008

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		the presence of this skew in the power on sequence.	
3.3.2	Start Up	Start Up shall (3.3.2.19) test to ensure that all four FCPs are synchronized.	SRS008, SRS010
3.3.2	Start Up	Unsynchronized processors shall (3.3.2.20) be excluded from the FCP configuration.	SRS010, SRS296
3.3.2	Start Up	During start up the FCP watchdog timer shall (3.3.2.45) be active.	SRS014, SRS292
3.3.2	Start Up	System Initialization shall (3.3.2.21) initiate execution of FTSS and X-38 application code.	SRS199
3.3.3	Vehicle/Mission Manager	The Mission Manager Template shall (3.3.3.1) provide a mechanism for (but not limited to) creating and controlling task execution, creating message queues and other interprocessor communication mechanisms, and provide on/off capability for processor resynchronization.	SRS215, SRS302
~~3.3.4~~	~~Scheduler~~	~~The Scheduler shall (3.3.4.1) support three rate groups:~~	~~SRS197, SRS195, SRS035~~
3.3.4	Scheduler	The Scheduler shall (3.3.4.1) support three rate groups: 50 Hz (minor frame), 10 Hz (medium frame), and 1 Hz (major frame).	SRS197, SRS195, SRS035
3.3.4	Scheduler	The FTSS software shall (3.3.4.18) take at	SRS024, SRS034

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		most 1 msec of a 50 Hz minor frame.	
3.3.4	Scheduler	The Scheduler shall (3.3.4.3) set the timer to a count down value so as to cause the next minor frame interrupt at 20 msec (+/- 330 usecs) from the previous interrupt congruently in all operational FCPs.	SRS035, SRS181
3.3.4	Scheduler	The FTSS software shall (3.3.4.19) provide an API call which provides the application program the minor frame number.	SRS278
3.3.4	Scheduler	Process scheduling shall (3.3.4.4) only be performed at certain controlled locations (synchronization points).	SRS022
3.3.4	Scheduler	The Scheduler shall (3.3.4.5) place processing time bounds on all rate groups to ensure that no rate group monopolizes the FCC's processor.	SRS028
3.3.4	Scheduler	It shall (3.3.4.6) be possible to reassign a task to a different rate group as a function of the mission mode.	SRS017, SRS018, SRS195, SRS196, SRS197, SRS198
3.3.4	Scheduler	Tasks within a rate group shall (3.3.4.7) be executed in the order in which the mission manager registers the tasks.	SRS022, SRS039, SRS037, SRS198
3.3.4	Scheduler	It shall (3.3.4.8) be possible to alter the execution sequence of tasks within a rate	SRS002, SRS017, SRS196, SRS018, SRS019, SRS020, SRS021, SRS197,

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		group as a function of mission mode.	SRS195, SRS198
3.3.4	Scheduler	Higher iteration tasks shall (3.3.4.9) have higher priority over lower iteration tasks.	SRS027
3.3.4	Scheduler	The Scheduler shall (3.3.4.12) detect 50 Hz, 10 Hz, and 1 Hz frame overruns at the next frame following the end of their respective rate boundaries.	SRS028
3.3.4	Scheduler	The Scheduler shall (3.3.4.13) attempt recovery from a frame overrun according to the following policy:	SRS030
3.3.4	Scheduler	if the scheduler determines that a task did not finish within its specified rate boundary, the scheduler shall (3.3.4.14) signal that a task overrun occurred.	SRS030
3.3.4	Scheduler	When the task restart begins, the FTSS shall (3.3.4.15) provide a mechanism to signal the task to execute its startup recovery actions, including updating the I-Load data and pre-stored last good data state.	SRS030
3.3.4	Scheduler	Following a task overrun, the scheduler shall (3.3.4.17) provide an application programmer's interface call which specifies which task was running within the rate group which has overrun.	SRS216
3.3.4	Scheduler	The FTSS software	SRS270

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		shall (3.3.4.20) provide a task deadline capability which allows an application to specify which minor frame that an application should start in and finish in.	
3.3.4	Scheduler	The Scheduler in the FTPP shall (3.3.4.16) keep all redundant copies of a process, which are executing in different computers, in synchronization.	SRS181
3.3.5	Fault Detection, Identification, and Recovery	The scope of FDIR shall (3.3.5.1) be limited to the hardware on the four FCP boards, the four MPCC/CTC boards, the five ICPs, and the five NEs.	SRS095, SRS096, SRS184, SRS235, SRS298, SRS299, SRS300, SRS304
3.3.5	Fault Detection, Identification, and Recovery	FDIR shall (3.3.5.2) receive this information and, after two consecutive missed “heartbeats,” conclude that the ICP is failed.	SRS097
3.3.5	Fault Detection, Identification, and Recovery	FDIR shall (3.3.5.3) report the total FCC status to the Vehicle/Mission Manager when requested to do so by the Vehicle/Mission Manager.	SRS098, SRS099
3.3.5	Fault Detection, Identification, and Recovery	The FDIR shall (3.3.5.4) execute CBIT during all operational phases.	SRS095
3.3.5	Fault Detection, Identification, and Recovery	The CBIT shall (3.3.5.5) be executed at a 50 Hz rate, after all 50 Hz flight critical operations are complete.	SRS093, SRS095, SRS034

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
3.3.5	Fault Detection, Identification, and Recovery	The CBIT, at a minimum, shall (3.3.5.6) include a "presence test" to ascertain that all FCP processors are synchronized and are at the same relative point in time in the current minor frame.	SRS093, SRS095, SRS184
3.3.5	Fault Detection, Identification, and Recovery	The presence test shall (3.3.5.7) also ascertain that all processors are executing the same 50 Hz, 10 Hz, and 1 Hz frames.	SRS184
3.3.5	Fault Detection, Identification, and Recovery	The CBIT shall (3.3.5.8) also arm and reset the hardware watchdog timer.	SRS014, SRS094
3.3.5	Fault Detection, Identification, and Recovery	The CBIT shall (3.3.5.10) be executed without interfering with the normal execution of the application tasks.	SRS034
3.3.5	Fault Detection, Identification, and Recovery	The FDIR shall (3.3.5.11) not take more than 2 msec per minor frame under nominal no-fault conditions.	SRS091
3.3.5	Fault Detection, Identification, and Recovery	The FDIR shall (3.3.5.12) not take more than 3 msec per minor frame while processing faults.	SRS183
3.3.5	Fault Detection, Identification, and Recovery	The FDIR shall (3.3.5.13) be able to discriminate between permanent and non-permanent faults.	SRS106, SRS110, SRS117, SRS204, SRS208, SRS209, SRS211, SRS282, SRS298
3.3.5	Fault Detection, Identification, and Recovery	The FDIR shall (3.3.5.14) reset and retry the failed entity, such as an FCP or an NE, to perform this	SRS106, SRS110, SRS117, SRS129 SRS204, SRS208, SRS209, SRS211, SRS282

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		discrimination.	
3.3.5	Fault Detection, Identification, and Recovery	To clear the failure, FDIR shall (3.3.5.15) request the Vehicle/Mission Manager to cycle power to that FCR.	SRS208, SRS209
3.3.5	Fault Detection, Identification, and Recovery	The FDIR shall (3.3.5.16) be able to identify a fault source, at least to an FCR.	SRS095, SRS184, SRS096, SRS097
3.3.5	Fault Detection, Identification, and Recovery	The FDIR shall (3.3.5.17) place all fault and recovery information in shared memory for inclusion in the frames that will be telemetred and recorded by the CTC.	SRS098, SRS044
3.3.5	Fault Detection, Identification, and Recovery	For the first permanent FCP failure, FDIR shall (3.3.5.30) degrade the redundancy level of the FCP from 4 to 3.	SRS106
3.3.5	Fault Detection, Identification, and Recovery	If a second permanent FCP failure occurs, then FDIR shall (3.3.5.31) degrade the redundancy level of the FCP from 3 to 2 and operate in a degraded triplex mode.	SRS282
3.3.5	Fault Detection, Identification, and Recovery	Additionally, FDIR shall (3.3.5.20) reinitialize and integrate an FCP if permitted by the Vehicle/Mission Manager.	SRS104, SRS110, SRS123, SRS124, SRS125, SRS126, SRS281, SRS302
3.3.5	Fault Detection, Identification, and Recovery	FTSS shall (3.3.5.35) notify the applications that memory re-alignment and re-integration of an FCP is going to occur in 1 second.	SRS271

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
3.3.5	Fault Detection, Identification, and Recovery	FTSS shall (3.3.5.36) wait for the ICP to signal that it has completed initialization before suspending the application for memory re-alignment.	SRS272
3.3.5	Fault Detection, Identification, and Recovery	The FCP watchdog timer shall (3.3.5.38) remain active during memory re-alignment.	SRS293, SRS294
3.3.5	Fault Detection, Identification, and Recovery	Reintegration of an FCP shall (3.3.5.21) be completed in at most 1.5 minutes.	SRS214
3.3.5	Fault Detection, Identification, and Recovery	It shall (3.3.5.22) be possible to perform voted VMEbus resets via the NEs.	SRS204
3.3.5	Fault Detection, Identification, and Recovery	For a permanent NE failure, FDIR shall (3.3.5.23) mask the failed NE.	SRS104, SRS245
3.3.5	Fault Detection, Identification, and Recovery	For a transient NE failure, FDIR shall (3.3.5.24) mask the failed NE.	SRS104, SRS106, SRS245, SRS282
3.3.5	Fault Detection, Identification, and Recovery	Additionally, FDIR shall (3.3.5.25) reinitialize and integrate the NE.	SRS104
~~3.3.5~~	~~Fault Detection, Identification, and Recovery~~	~~FTSS shall (3.3.5.34) provide an API call which allows the application to notify FTSS that an FCP, ICP, or NE is intentionally being powered down.~~	~~SRS274~~
3.3.5	Fault Detection, Identification, and Recovery	Intentional powering down of an FCP, ICP, or NE shall (3.3.5.32) not be classified as a fault.	SRS285, SRS274, SRS128
3.3.5	Fault Detection, Identification, and Recovery	FTSS shall (3.3.5.34) provide an API call which allows the application to notify	SRS274

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		FTSS that an FCP, ICP, or NE is intentionally being powered down.	
3.3.5	Fault Detection, Identification, and Recovery	FTSS shall (3.3.5.37) provide an API call which allows the application to take an FCR out of the permanently failed state and place it back in the initial recovery state.	SRS285
3.3.5	Fault Detection, Identification, and Recovery	A failed FCP or NE shall (3.3.5.27) be masked within three minor frames of fault detection and isolation.	SRS109
3.3.5	Fault Detection, Identification, and Recovery	The FTSS FDIR shall (3.3.5.28) exchange the status information of detected faults in the FCP, ICP, NE, and MPCC/CTC hardware with the NASA provided software.	SRS044, SRS098
3.3.5	Fault Detection, Identification, and Recovery	The FTPP system shall (3.3.5.29) perform the "FTPP Failure Response/Recovery Mechanisms"as listed in the following matrix and notes of interest.	SRS100, SRS104, SRS242, SRS222, SRS208, SRS209, SRS211, SRS245, SRS283, SRS284, SRS298, SRS299, SRS300, SRS304
3.3.6	Communications	Synchronous communication shall (3.3.6.1) be in the form of messages enqueued for transmission at the start of the next rate group frame and dequeued for reading by the recipient task within the next rate group frame after it is received.	SRS047, SRS052, SRS063, SRS064
3.3.6	Communications	A transmit packet	SRS053, SRS054,

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		queue and a receive packet queue shall (3.3.6.7) be maintained for each task or Communication ID (CID).	SRS055, SRS059, SRS062, SRS066
3.3.6	Communications	Access to the transmit queues shall (3.3.6.8) be controlled within the communication service primitives .	SRS062, SRS066
3.3.6	Communications	Message passing communications primitives shall (3.3.6.9) be provided for task-to-task communication as well as for broadcast to all processors.	SRS047, SRS048, SRS049, SRS051 SRS052, SRS062, SRS069
3.3.6	Communications	Broadcast primitives shall (3.3.6.10) not be available on ICPs.	SRS064, SRS070
3.3.6	Communications	For the highest rate group tasks (i.e., tasks that can not be preempted), Immediate Message Services shall (3.3.6.11) also be provided.	SRS048, SRS050, SRS067, SRS068, SRS069, SRS070, SRS073
3.3.6	Communications	A version of the Immediate Message Services shall (3.3.6.12) be provided to the ICPs that allows Class 2 writes to NEs and Class 1 reads from NEs.	SRS226, SRS227, SRS228, SRS229, SRS230, SRS303
3.3.6	Communications	Communications services shall (3.3.6.13) provide a version of Immediate Message Services between rate groups within the FCP that bypasses the NE and that can be used to control and monitor	SRS051

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		inter-rate group communications.	
3.3.6	Communications	Communication services shall (3.3.6.14) provide the capability for a “helper” task to be created to run in the 50 Hz rate group, but running in specific minor cycles (every 5th or every 50th) to provide data from the ICP to the lower rate tasks.	SRS042
3.3.7	System Loader	The procedures used to build the executable FTSS software using a cross-compiler and linker, down-load the image to the target processors, and burn the load image into the Radstone PowerPC flash RAM shall (3.3.7.1) be documented in the release notes that accompany Engineering Releases of the FTSS software and in the Software Users Manual.	NA
3.3.7	System Loader	Any makefiles or other automated scripts that support the build, down-load, and flash programming processes shall (3.3.7.2) be delivered with the software to NASA.	NA
3.3.8	Memory Management	For each mission mode, the congruent and non-congruent memory boundaries shall (3.3.8.1) be known and fixed.	SRS217
3.3.8	Memory Management	The Memory	SRS043

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		Management software shall (3.3.8.2) periodically "scrub" volatile and read/write memory in the FCP.	
3.3.8	Memory Management	It shall (3.3.8.3) not be necessary to scrub memory that is not used by the flight software.	SRS043
3.3.8	Memory Management	It shall (3.3.8.16) not be necessary to scrub that area used to store telemetry data.	SRS275
3.3.8	Memory Management	Memory scrubbing shall (3.3.8.5) be executed without interfering with normal execution of applications tasks.	SRS187
3.3.8	Memory Management	The memory scrubbing software shall (3.3.8.6) be capable of scrubbing 10 Megabytes in 8 minutes.	SRS187
3.3.8	Memory Management	The RAM scrub software shall (3.3.8.15) at most use 1% of an FCP CPU duty cycle.	SRS187
3.3.8	Memory Management	Even though memory scrubbing is performed locally and the errors would not be congruent, the recording of errors shall (3.3.8.10) be congruent.	SRS044
3.3.8	Memory Management	To support reintegrating a desynchronized channel, as specified in the FDIR requirements, the Memory Management software shall (3.3.8.11) "re-align" all	SRS045, SRS126, SRS186, SRS200

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		of the volatile and read/write congruent memory, registers, timers and other locations that fit the description of "volatile and read/write congruent locations".	
3.3.8	Memory Management	The re-align function shall (3.3.8.12) write the voted value from the good channels into the target channel.	SRS186, SRS281
3.3.8	Memory Management	The re-align function shall (3.3.8.13) be allowed only when permitted by the Vehicle/Mission Manager.	SRS110, SRS117, SRS125, SRS302
3.3.8	Memory Management	Memory Management software shall (3.3.8.14) include (but not be limited to) memory scrubbing and memory realignment.	SRS043, SRS044, SRS046, SRS217, SRS045, SRS186, SRS200, SRS203, SRS187
3.3.9	Memory Protection	Memory shall (3.3.9.1) be categorized into congruent (identical data) and non-congruent (data that is not identical) memory.	SRS046
3.3.10	Time Management	Time Management shall (3.3.10.4) provide MET.	SRS142
3.3.10	Time Management	The MET shall (3.3.10.5) be initialized to zero at the first 50 Hz frame.	SRS165
3.3.10	Time Management	The MET shall (3.3.10.6) measure real-time from this event with an accuracy of at most 50 Parts Per Million (PPM).	SRS218
3.3.10	Time Management	The MET shall (3.3.10.7) have a	SRS142

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		resolution of 20 msec. for 50 Hz tasks, 100 msec for 10 Hz tasks, and 1 second for 1 Hz tasks.	
3.3.10	Time Management	The MET shall (3.3.10.8) be able to increment to at least 30 days without rolling over.	SRS144
3.3.10	Time Management	The MET shall (3.3.10.9) be congruent across all FCP members.	SRS142
3.3.10	Time Management	Following a processor recovery, during which time is frozen, the FTSS software shall (3.3.10.10) account for the frozen time and update MET to its proper value.	SRS218
3.3.10	Time Management	Time Management shall (3.3.10.11) provide SEP.	SRS142
3.3.10	Time Management	Time Management shall (3.3.10.12) initialize SEP to zero within one minor cycle of the time when the vehicle/mission manager software has notified the FTSS software that the X-38 vehicle is released from the Space Shuttle Remote Manipulator System.	SRS161
3.3.10	Time Management	The SEP shall (3.3.10.13) measure real-time from this event with an accuracy of at most 50 Parts Per Million (PPM).	SRS219
3.3.10	Time Management	The SEP shall (3.3.10.14) have a resolution of 20 msec.	SRS142

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		for 50 Hz tasks, 100 msec for 10 Hz tasks, and 1 second for 1 Hz tasks.	
3.3.10	Time Management	The SEP shall (3.3.10.15) be able to increment to at least 1 day without rolling over.	SRS145
3.3.10	Time Management	The SEP shall (3.3.10.16) be congruent across all FCP members.	SRS142
3.3.10	Time Management	Following a processor recovery, during which time is frozen, the FTSS software shall (3.3.10.17) account for the frozen time and update SEP to its proper value.	SRS219
3.3.10	Time Management	If the SEP API call is made prior to actual separation, the call shall (3.3.10.27) return zero (0).	SRS161
3.3.10	Time Management	The Time Services, if dealing with the year designation, shall (3.3.10.21) be Year 2000-compliant.	No year designation used in any requirement
3.3.10	Time Management	Time Management shall (3.3.10.23) provide a utility timer.	SRS246, SRS248
3.3.10	Time Management	The utility timer shall (3.3.10.24) be available via an FTSS API call(s).	SRS246
3.3.10	Time Management	The utility timer shall (3.3.10.25) have a resolution of 60.6 nanoseconds.	SRS256
3.3.10	Time Management	The utility timer shall (3.3.10.26) have an accuracy of at most 50 PPM.	SRS247
3.3.11	Input/Output Services	Load modules of the four FCPs shall	SRS166

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		(3.3.11.1) be identical.	
3.3.11	Input/Output Services	Control flow of the four FCPs shall (3.3.11.2) be similar, if not identical.	SRS008, SRS011, SRS181, SRS191, SRS053, SRS054, SRS095, SRS184, SRS123, SRS125, SRS126, SRS045, SRS186, SRS200, SRS166, SRS168
3.3.11	Input/Output Services	Asymmetric I/O calls shall (3.3.11.3) not be allowed to induce a large enough skew to force the FCPs to desynchronize.	SRS168
3.3.11	Input/Output Services	A subset of FTSS shall (3.3.11.7) reside on the ICP.	SRS225, SRS226, SRS227, SRS228, SRS229, SRS230
3.3.11	Input/Output Services	FTSS shall (3.3.11.8) provide an API call which allows the application to specify which MPCC channels in a C&T FCR should be used for telemetry and/or command reception.	SRS286
3.3.12	Exception Handling	Upon occurrence of an exception, the FTPP system shall (3.3.12.1) log the error and include all context data relevant to the exception e.g. the contents of the Machine State Register (MSR) and the machine status Save/Restore Registers (SRR0 & SRR1).	SRS172
3.3.12	Exception Handling	The error type and its context data shall (3.3.12.4) be made available to the application via an API call.	SRS172
3.3.12	Exception Handling	For software	SRS031

FTPP Section #	**FTPP Section Name**	**FTPP Requirement**	**SRS Req #s**
		exceptions, the FTPP system shall (3.3.12.2) then transfer control to a user specified exception handling routine, if one is provided.	
3.3.12	Exception Handling	For hardware exceptions, the FTPP system shall (3.3.12.5) "handle" the exception by making the error and its context data available to the application and then returning from the exception handler.	SRS276
3.3.12	Exception Handling	For reserved exceptions, the FTPP system shall (3.3.12.6) "handle" the exception by making the error and its context data available to the application and then returning from the exception handler.	SRS276
3.3.12	Exception Handling	Finally, for software exceptions only, the FTPP system shall (3.3.12.3) then "jump back" to the initialization point for the offending task.	SRS173
3.3.12	Exception Handling	If the exception occurs within the FTSS software, the FTPP system shall (3.3.12.7) "jump" back to the beginning of the task, skip all initialization code, and begin processing the task's code again.	SRS277, SRS301
3.3.13	Application Interface	An application programming interface shall (3.3.13.3) be documented in a FTSS API document.	SRS164

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
3.3.14	NEFU	The presence, or absence of, an NEFU ICP shall (3.3.14.1) not impact the FTSS software (i.e., the FTSS ICP load will not be different).	SRS220
3.3.15	Power Down	FTSS services shall (3.3.15) provide an API call which provides the capability to close and delete all communication mechanisms delete all rate groups, and suspend and delete all tasks.	SRS249
~~3.4.1.1~~	~~FCP-ICP Communication Architecture~~	~~The FCP-ICP communications shall (3.4.1.1.1) provide the following capabilities (these are written from the viewpoint of the FCP):~~ ~~1. Signal the start of the 50 Hz rate group in the ICP.~~ ~~2. Synchronize frames across the four ICPs.~~ ~~3. Receive congruent sensor data from ICPs.~~ ~~4. Send voted actuator and other output device commands to ICPs.~~ ~~5. Receive health and status of the ICPs and all the FCC hardware for which the ICPs are responsible.~~ ~~6. Provide the ICPs with current minor frame number, X-38 flight phase/segment number, vehicle mode number, MET, and SEP.~~ ~~7. Provide th~~	~~SRS025, SRS032, SRS033, SRS048, SRS097, SRS231, SRS232, SRS233, SRS295~~

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
3.4.1.1	FCP-ICP Communication Architecture	The FCP-ICP communications shall (3.4.1.1.1) provide the following capabilities (these are written from the viewpoint of the FCP): 1. Signal the start of the 50 Hz rate group in the ICP. 2. Synchronize frames across the four ICPs. 3. Receive congruent sensor data from ICPs. 4. Send voted actuator and other output device commands to ICPs. 5. Receive health and status of the ICPs and all the FCC hardware for which the ICPs are responsible. 6. Provide the ICPs with current minor frame number, X-38 flight phase/segment number, vehicle mode number, MET, and SEP. 7. Provide the ICPs with notification of FCP memory alignment two minor frames prior to the start of the alignment.	SRS025, SRS032, SRS033, SRS048, SRS097, SRS231, SRS232, SRS233, SRS295
3.4.1.2	FCP-ICP Communication Requirements	As part of start-up or after recovering from a transient fault: after completing IBIT, the FCP shall (3.4.1.2.1) wait 15 seconds for the ICP to initialize all of its non-Radstone VME slave boards and its NE interface	SRS297
3.4.1.2	FCP-ICP Communication Requirements	then the FCP VG shall (3.4.1.2.2) send the FCP VG Ready Signal to the ICPs to indicate that the FCP VG is	SRS221

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		ready to begin FCP-ICP communications.	
3.4.1.2	FCP-ICP Communication Requirements	To permit these task and pipe initializations in the ICPs, the FCP VG shall (3.4.1.2.8) wait at least 2.5 seconds for the ICP Ready Signals after the FCP VG has been notified that the ICPs received the FCP VG Ready Signals.	SRS189
3.4.1.2	FCP-ICP Communication Requirements	The FCP shall (3.4.1.2.5) signal the start of each minor frame in all ICPs by means of a VMEbus IRQ5 interrupt.	SRS032
3.4.1.2	FCP-ICP Communication Requirements	The interrupts across all channels shall (3.4.1.2.6) have a skew no greater than 330 microseconds.	SRS191
3.4.1.2	FCP-ICP Communication Requirements	Each interrupt shall (3.4.1.2.7) be preceded by the FCP writing the information listed in 3.4.1.1.1-6 to a shared memory block over the VME backplane bus to its counterpart ICP.	SRS033
~~3.4.2~~	~~FCP-CTC Communications~~	~~The FTSS software shall (3.4.2.2) provide to the telemetry program the FTPP telemetry data which consists of the following elements:~~	~~SRS029, SRS044, SRS098~~
3.4.2	FCP-CTC Communications	The FTSS software shall (3.4.2.2) provide to the telemetry program the FTPP telemetry data which consists of the following elements: a. FCP status, b. NE status, c. # Transient	SRS029, SRS044, SRS098

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		Errors, d. Transient recovery attempts, e. Frame overruns, f. ICP status.	
3.4.2	FCP-CTC Communications	The data mentioned in requirement 3.4.2.2 and any other Draper-provided telemetry data shall (3.4.2.9) fit within Draper's allocated telemetry budget of 5000 bits/sec.	SRS250
3.4.2	FCP-CTC Communications	In addition, the FTSS software shall (3.4.2.11) provide up to 600 bits of start-up data that indicates the state of the FTPP system during start-up.	SRS280
3.4.2	FCP-CTC Communications	Once every medium frame, the FTSS software shall (3.4.2.3) accept a pointer from the telemetry program to the buffer space containing the telemetry data.	SRS148, SRS151
3.4.2	FCP-CTC Communications	The FTSS software shall (3.4.2.4) move this buffer to the MPCC/CTC over the VMEbus.	SRS149, SRS150
3.4.2	FCP-CTC Communications	The FTSS software shall (3.4.2.10) use no more than 5.2 milliseconds of FCP processing time to move the telemetry data to the MPCC/CTC board and complete communication and error handling for the MPCC/CTC board.	SRS257
3.4.2	FCP-CTC Communications	The FTSS software shall (3.4.2.5) receive	SRS152

FTPP Section #	FTPP Section Name	FTPP Requirement	SRS Req #s
		telemetry commands from both CTCs via the MPCC/CTC once every medium frame.	
3.4.2	FCP-CTC Communications	The FTSS software shall (3.4.2.6) congruently decide which FCP channel should be the source of CTC data.	SRS222
3.4.2	FCP-CTC Communications	This decision shall (3.4.2.7) be made based on the health and status of all the physical links from the FCP to the CTC.	SRS222
3.4.2	FCP-CTC Communications	The FTSS software shall (3.4.2.8) deliver to the requisite NASA application program commands received from both CTCs.	SRS153, SRS156
3.5	Miscellaneous Other Requirements	Draper shall (3.5.5) not violate the 100 microseconds requirement.	SRS223
3.5	Miscellaneous Other Requirements	Draper shall (3.5.6) deliver FTSS engineering release version 5/6, and any subsequent versions of the FTSS software, only after the release has been proven to work under Tornado 2 for the NT environment.	SRS253
3.5	Miscellaneous Other Requirements	Draper shall (3.5.7) deliver FTSS engineering release version 5/6, and any subsequent versions of the FTSS software, via CD-ROM.	SRS252